HiFRAP installation and use instructions

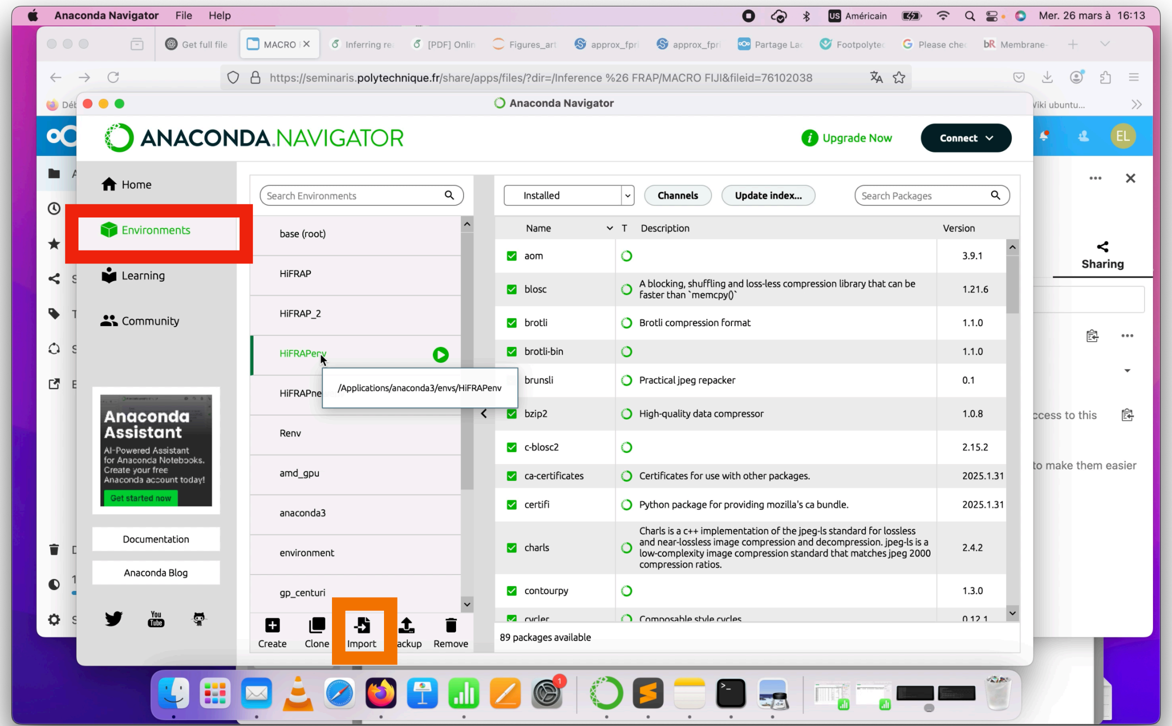
1. Download HiFRAP

1. Download the latest version of HiFRAP (e.g. HiFRAP_v001).
2. Unzip the downloaded file and ensure that the folder contains a folder named **HiFRAP**.

2. Install HiFRAP environment using Anaconda

1. Open **Anaconda Navigator**. If you have not installed it on your computer, go to **https://www.anaconda.com/products/distribution** and download the appropriate version for your operating system.
2. Go to **Environments** (left sidebar), as indicated by the red box in Figure 1.
3. Click **Import** in the bottom bar, as indicated by the orange box in Figure 1.
4. In the file dialog box, navigate to the **HiFRAP** folder that you have extracted earlier and go to the **env** folder. Select the **HiFRAPenv.yaml** file to import it.
5. Wait for the installation to complete (this may take several minutes).

Figure 1



6. Once the HiFRAP environment is installed, **visualise its path** in Anaconda Navigator by moving the cursor on the HiFRAPenv bar as shown in Figure 1 (typical path: Applications/ Anaconda3/HiFRAP_env). Alternatively, click on the arrow in the green circle and select 'Open Terminal'. The path appears after 'conda activate' in the command that is executed.

7. **Keep a note** of this path for later use.

8. You may close the Anaconda Navigator; tt will not be used anymore by HiFRAP.

2. Install HiFRAP in Fiji

Step 1: Copy HiFRAP to the Fiji plugins folder

1. Navigate to the extracted **HiFRAP** folder.

2. Copy the entire **HiFRAP** folder.

3. Locate your **Fiji** installation directory.

4. Open the **plugins** folder within Fiji (with Fiji closed).

5. Paste the **HiFRAP** folder inside the **plugins** directory.

Step 2: Launch Fiji and load HiFRAP

1. Open **Fiji**.

2. Load an image file for analysis. Two sample .tiff file (1D and 2D FRAP experiments) are included in the **Examples** folder inside **HiFRAP** folder.

3. Locate and select **HiFRAP** from the list of **Plugins** in the Fiji toolbar. If HiFRAP does not appear, verify that the folder was correctly placed in the **plugins** directory or run install plugin in the plugins bar.

3. Run HiFRAP analysis

Step 1: Input parameters

Once HiFRAP is launched, a dialog box prompts you for the input parameters (Figure 2A):

1. **HiFRAP environment path**: Enter the Anaconda environment path noted earlier.

2. **Output directory**: Choose where you want to save results.

3. **Experiment name**: Assign a label to your analysis. Do not use spaces in the name (use underscores _ instead).

Figure 2A

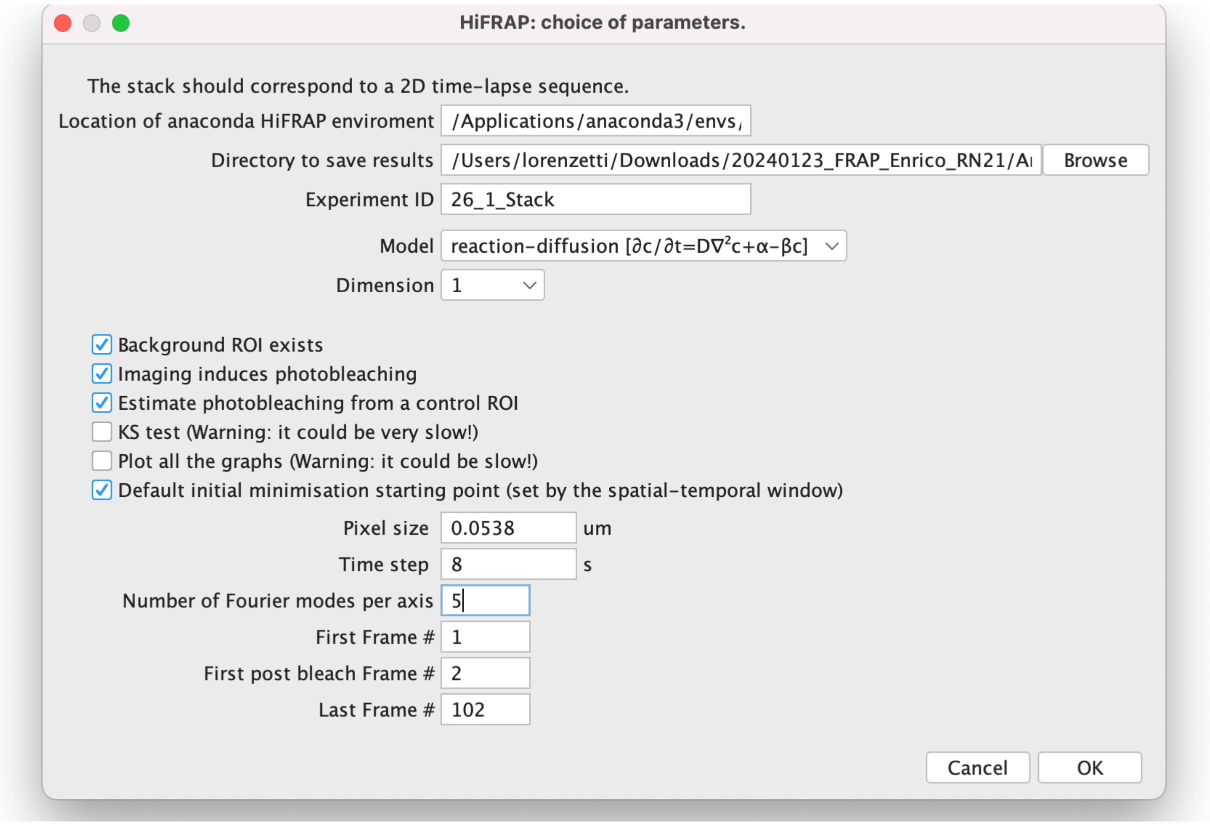
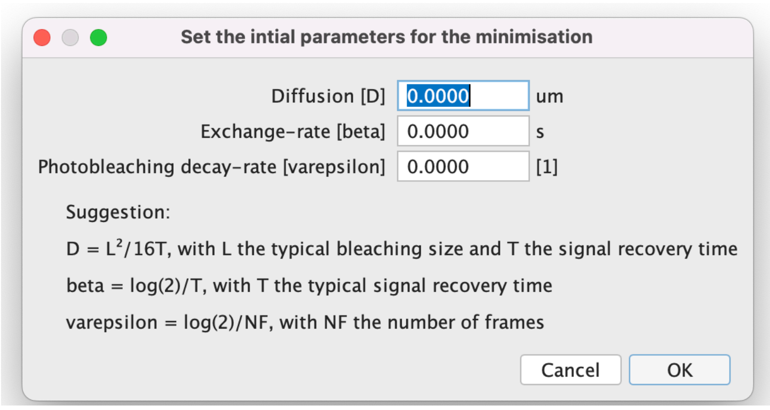


Figure 2B



4. **Select model type**:

- Diffusion, described by the partial differential equation (PDE) $\partial_t c = D\nabla^2 c$.
- Reaction-diffusion, described by the PDE $\partial_t c = D\nabla^2 c + \alpha - \beta c$.
- Reaction, described by the PDE $\partial_t c = \alpha - \beta c$.

5. **Dimension:**

- 1D for FRAP experiments on a (possibly curved) line.

- 2D for FRAP experiments on a flat surface.

6. **Image processing settings**:

   - **Background ROI exists**: Tick the checkbox if a background (sample-free) area is available.

   - **Imaging-induces photobleaching**: Tick the checkbox if imaging induces gradual photobleaching.

   - **Estimate photobleaching from a control ROI**: Tick the checkbox if applicable.

6. **KS test**:

   - Tick the checkbox to also evaluate fit quality with the Kolmogrov-Smirnov test. This test is very slow.

7. **Plot all the graphs**:

   - Tick the checkbox to generate plots of curve fits and cost function. This may slow down computation.

8. **Default initial minimisation starting point**:

   - Tick the checkbox to use the default values of initial parameters for diffusion and/or exchange-rate and/or photobleaching decay-rate based on the spatial-temporal window of the analysed time-lapse video. If not, you may insert these values manually as in figure 2B.

8. **Pixel size and time step**:

   - Input a specific value (e.g., 0.00538) for the pixel size and time step (e.g., 8) .

9. **Frame selection**:

   - First frame, including photobleaching frame, to analyse (e.g 1).

   - First post-bleaching frame (e.g 2).

   - Last frame to analyse.

10. **Fourier modes**:

    - Use an **odd integer (e.g., 5, 7, 9)**. Start with **5** for fast computation.

Step 2: Select Regions of Interest (ROI)

In 2D:

1. Select a **sample-free area** for **background estimation** as in figure 3A (appears only if corresponding checkbox ticked earlier).

2. **Select a control area** that has not been FRAPped to estimate photobleaching during imaging as in Figure 3B (appears only if corresponding checkbox ticked earlier).

3. Select the **FRAPped region and its surroundings** as in Figure 3C.

Figure 3A: Background

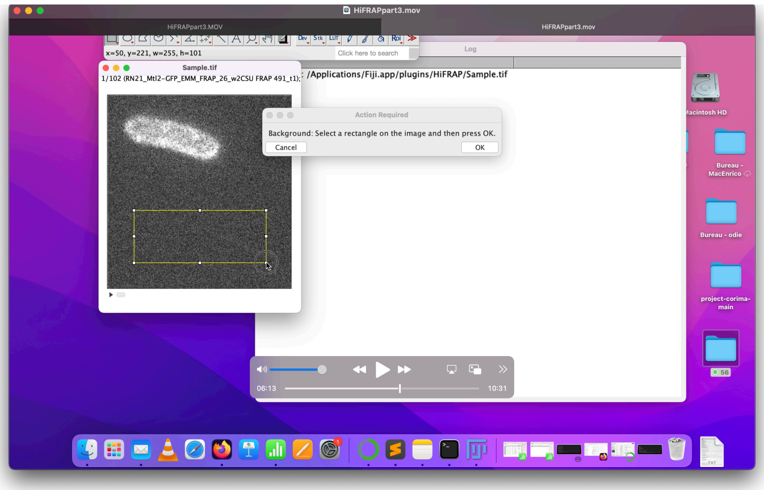


Figure 3B: Control area

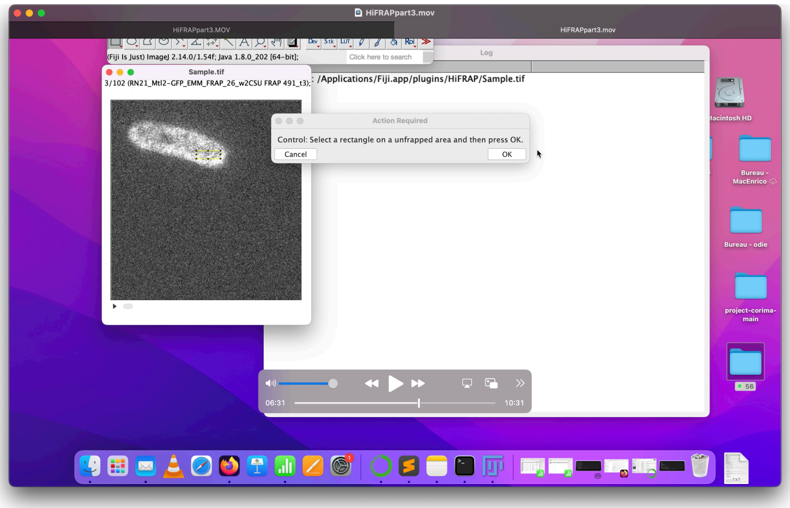
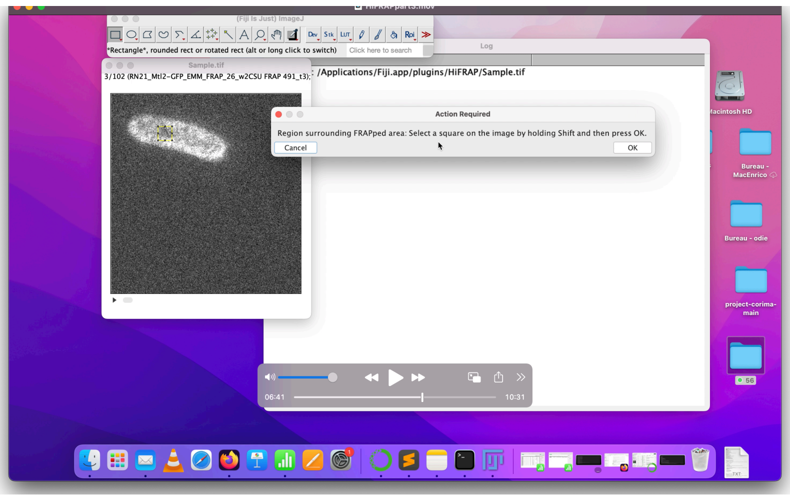


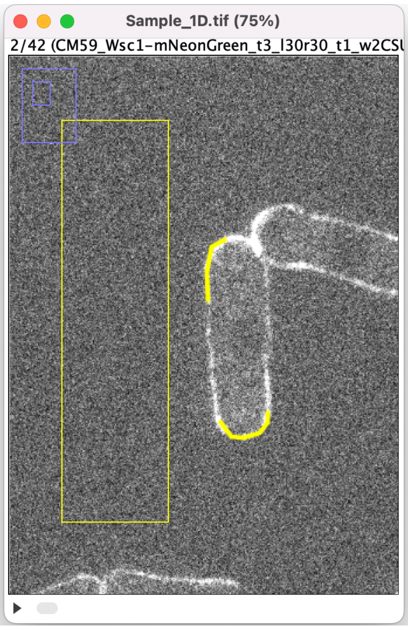
Figure 3C: FRAP ROI



In 1D:

Analogous to 2D case except that you need to select polylines (of thickness to be specified) instead of rectangles/squares for the FRAP and control ROIs, as shown in Figure 3D.

Figure 3D: 1D ROIs



Step 3: Run Analysis

1. Click on Ok to run the Analysis.
2. Wait for the analysis to be finished (several seconds).

Step 4: Visualise the results

1. If successful, the program will generate various plots (provided you ticked the corresponding checkbox):
   - **Cost Function Plot (Linear & Log Scale)**: Ensure the minima is well centred.
   - **Fit of Time-lapse Video**. In 2D the best fit is plotted for 4 frames occurring at 0, T/3, 2T/3 and T with T the total duration of the video from the initial postbleach frame. In 1D the best fit is represented with a kymograph (space vs time).
2. Close the plots to complete the analysis. These plots are automatically saved in the output folder with a name based on the experiment name and model used for analysis. Ex: Experiment_1_diff_data_fit.pdf
3. Next, all the results can be found in:
   - **Fiji Log File** (Viewable within Fiji).

- **Results.csv** file in the output directory.

Full output description

1. **Experiment name**
   The name assigned to the experiment. By default: the last two directories of the image path + model + .tiff title.

2. **Dimension**
   Indicates whether the experiment was done in 1D or 2D.

3. **Model**
   Type of model used: "diff" (diffusion only), "reac-diff" (reaction-diffusion), or "reac" (reaction only).

4. **Background**
   Indicates whether a background ROI (Region of Interest) was defined: 1 = yes, 0 = no.

5. **Control**
   Indicates whether a control ROI was defined: 1 = yes, 0 = no.

6. **Photobleaching**
   If 1, photobleaching during acquisition was considered to be significant. 0 otherwise.

7. **Time step** - Δt
   Time between two image frames (in seconds).

8. **Pixel size** - Δx
   Size of one pixel in micrometers (μm).

9. **First Frame**
   Frame number where analysis begins, includes pre-bleach frames.

10. **Postbleach frame**
    Number of the frame that follows FRAPping.

11. **Last frame**
    Frame number where the analysis ends.

12. **Number of pixels per frame** - n_x
    Total number of pixels analysed per frame per axis.

13. **Background coordinates**
    Coordinates of the background ROI in the format: (x_min, y_min, x_max, y_max).

14. **Control coordinates**
    Coordinates of the control ROI: (x_min, y_min, x_max, y_max) — for 2D analysis.

15. **ROI coordinates**
    Coordinates of the bleached (FRAP) ROI: (x_min, y_min, x_max, y_max) — for 2D analysis.

16. **Polyline ctrl [x,y]**
X and Y coordinates of the control ROI drawn with a polyline — for 1D analysis.

17. **Polyline frap [x,y]**
X and Y coordinates of the bleached ROI drawn with a polyline — for 1D analysis.

18. **Line thickness**
Thickness of the polylines — for 1D analysis.

19. **nqc** - n_q
Number of Fourier modes per axis used for compression.

20. **D_in** - D_{in}
Initial guess for the diffusion coefficient for the cost minimisation.

21. **beta_in** - β_{in}
Initial guess for the exchange-rate for the cost minimisation.

22. **varepsilon_in** - $\varepsilon_{\mathrm{in}}$
Initial guess for the photobleaching decay-rate for the cost minimisation.

23. **D_est** - D_{est}
Estimated diffusion coefficient.

24. **err_D** - σ_D
Error associated with the estimated diffusion coefficient.

25. **alpha_est** - α_{est}
Estimated alpha parameter (typically associated with reaction rate) in the A.U. of the signal.

26. **err_alpha** - σ_α
Error on the estimated alpha value in the A.U. of the signal.

27. **beta_est** - β_{est}
Estimated exchange rate (or dissociation rate) in seconds.

28. **err_beta** - σ_β
Error on the estimated beta value in seconds.

29. **varepsilon** - ε
Estimated photobleaching correction factor.

30. **err_varepsilon -** σ_ε
Error on the estimated photobleaching factor.

31. **bg_est** - $I^{\mathrm{BG}}_{\mathrm{est}}$
Estimated background intensity from model fitting (A.U.).

32. **err_bg** - $\sigma_{I_{\mathrm{BG}}}$
Error on the estimated background (A.U.).

33. **R^2_adj** - R^2_{adj}
Value of the adjusted coefficient of determination indicating the quality of the fit.

34. **kol_pvalue**
P-value from the Kolmogorov-Smirnov test to evaluate the goodness of fit.

35. **varepsilon0** - ε_0
Initial photobleaching decay rate estimated from the control area.

36. **bg0** - I_0^{BG}
Background intensity estimated directly from the background ROI (A.U.).

37. **cs** - c_s
Stationary (pre-bleach) concentration level (A.U.).

38. **covM**
Covariance matrix used to store all estimated errors.